A Novel Microcontroller-Based Sensorless Brushless DC (BLDC) Motor Drive for Automotive Fuel Pumps

Jianwen Shao, *Member, IEEE,*, Dennis Nolan, Maxime Teissier, and David Swanson

***Abstract*—This paper presents a novel back-electromotive-force (EMF) detection method for sensorless brushless dc (BLDC) motor drive systems. By this method, a true back-EMF signal can be directly extracted for each phase without sensing the neutral point of the motor. The method proposed is not sensitive to switching noise and requires no filtering. Good motor performance is achieved over a wide speed range as well. This novel sensing scheme is implemented into a hardware macro cell inside a mixed-signal microcontroller. The proposed microcontroller-based sensorless BLDC drive system has been successfully applied to automotive fuel-pump applications, which require high reliability and intelligence at a low cost.**

***Index Terms*—Back-electromotive-force (EMF) detection, fuel pumps, microcontroller, sensorless brushless dc (BLDC) drives.**

I. Introduction

THE brushless dc (BLDC) motor is well suited for automotive returnless fuel pump applications today. They are inherently more reliable, more efficient, and with current electronics technology, more cost effective than the standard brush-type fuel-pump motor and controller. In a returnless fuel system the fuel pump speed is adjusted to maintain constant fuel pressure over the fuel demand/load range. This requires some level of feedback and control. This control is typically performed by a microcontroller. Since a microcontroller is already in use, incorporating brushless commutation to the mix is incremental in cost. Over the last decade, there has been a steady improvement in electronics, control algorithms, and motor technologies. This improvement brings greater efficiencies in materials and functionality to the point of making the BLDC a preferred solution in not just automotive fuel pumps but also in a broad range of applications using adjustable speed motors.

Commutation sequencing can be determined in several ways. The most cost efficient in wye wound BLDC motors is sensing the back-electromotive-force (EMF) of an unused phase. In a typical three-phase wye wound BLDC motor utilizing six-step commutation, current flows in only two phases at any one time. This leaves the third phase available for commutation timing by sensing back EMF. Commutation timing is determined when the unused phase back EMF reaches its zero potential point. This is called the "zero crossing." The conventional and most commonly used method of sensing back EMF for this purpose requires a virtual neutral point that will, in theory, be at the same potential as the center of a wye-wound motor (see Fig. 2). The sensed back EMF is then the difference between the virtual neutral and the voltage at the floating terminal [1]–[3].

To control motor current/speed and keep system efficiency high the three phase inverter driving the motor is pulsewidth modulated. However, this pulsewidth-modulated signal is superimposed on the neutral voltage point. This induces a large amount of electrical noise on the sensed signal as well. To sense this properly requires both filtering and attenuation. Filtering is required to remove unwanted noise. The attenuation is required to bring the signal down to the allowable common mode range of the sensing circuit. The result is poor signal to noise ratio at low speeds, such as at startup, and significant commutation delays at high speeds. Consequently, this method tends to have a narrow speed range and poor startup characteristics. For fuel-pump applications reliable and fast startup is paramount.

To reduce the switching noise, the back-EMF integration [4] and third harmonic voltage integration [5] were introduced. The integration approach has the advantage of reduced switching noise sensitivity. However, they still have the accuracy problem at low speed. An indirect sensing of phase back-EMF zero crossing by detecting the conducting state of the freewheeling diodes in the unexcited phase was presented in [6]. The implementation of this method is complicated and costly, while the low-speed operation continues to be a problem.

This paper presents a novel back-EMF sensing method, which requires neither a manufactured neutral voltage nor a great amount of filtering. The true back-EMF zero crossing point can be extracted directly from the motor terminal voltage by properly choosing the pulsewidth-modualtion (PWM) and sensing strategy. The resulting feedback signal is not attenuated or filtered providing a timely signal with a very good signal/noise ratio. As a result this sensorless BLDC driver can provide a much wider speed range, from startup to full speed, than the conventional approach.

In recent years, with the development of mixed-signal integrated circuit technology, many system-on-chip (SOC) devices have become available. High-throughput microcontrollers

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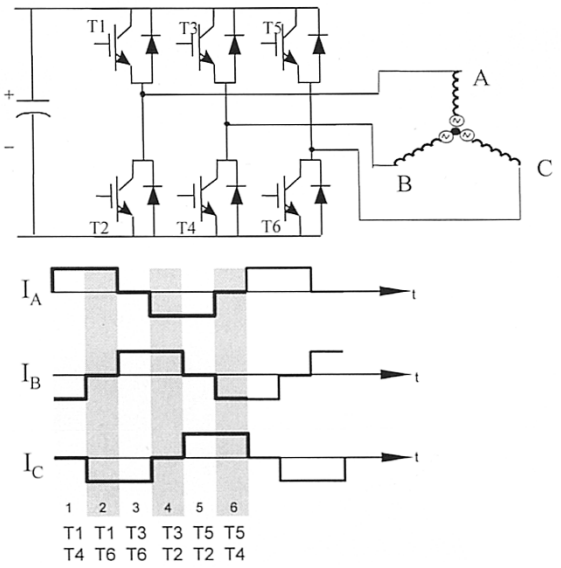


Fig. 1. Inverter configuration and current commutation sequence for BLDC motor.

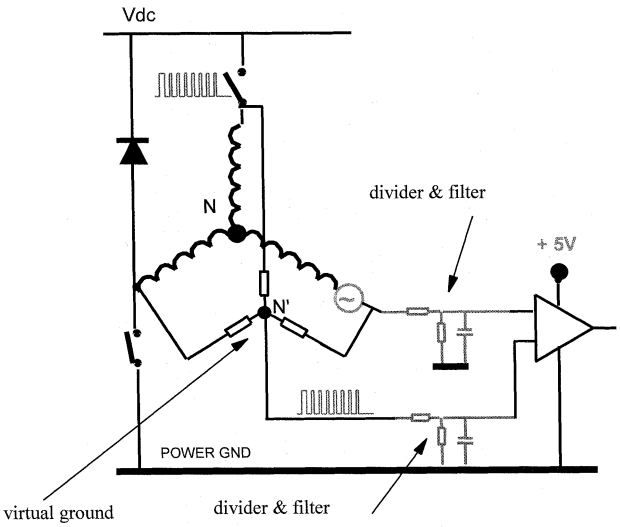


Fig. 2. Back-EMF sensing based on virtual neutral point.

with imbedded programmable memory as well as other precision analog and digital peripherals can be incorporated into a single integrated circuit. SOC devices have many advantages, including lower overall system cost and reduced board space, as well as superior system performance and reliability. A dedicated sensorless BLDC controller, implementing a back-EMF zero-crossing detection circuit as one of its peripherals, was therefore developed.

The proposed microcontroller-based sensorless BLDC drive system has been successfully applied to automotive returnless fuel pump applications. A brush type fuel pump motor is designed to last 6000 h. In certain fleet vehicles this can be expended in less than 1 year. A BLDC motor life span is typically around 15 000 h, extending the life of the motor by almost three times. Once a microcontroller is used to perform the brushless commutation other features can be incorporated into the application. Features such as electronic returnless fuel system control, fuel level processing, and fuel tank pressure sensing can be incorporated. These added features simplify the vehicle systems as well as drive overall system cost down.

II. Sensorless BLDC Motor Control

Typically, a BLDC motor is wound in a three-phase wye configuration. This configuration connects one end of each phase together to make a center point of a "Y" or the motor neutral point. This is then driven by a three-phase inverter with what is called six-step commutation. At any step, only two of the three phases are conducting current where current flows into one phase and then out another. For example, when phase *A* and phase *B* conduct current, phase *C* is floating. A transition from one step to another step is called commutation. So, totally, there are six steps in one cycle. As shown in Fig. 1, the first step is *AB* (phase *A* and *B* conducting current), then to *AC*, to *BC*, to *BA*, to *CA*, to *CB* and then the pattern is repeated. In order to produce maximum torque, the inverter should be commutated every 60 electrical degrees so that current is in phase with the back EMF. The conducting interval for each phase is 120 electrical degrees, or two steps. The commutation timing is determined by the rotor position, which can be determined every 60 electrical degrees by detecting when the back EMF on the floating phase crosses the zero potential point, or "zero crossing." Fig. 1 shows the typical inverter configuration and current commutation sequence.

As mentioned above, since only two phases are conducting current at any step, one phase is available to measure back EMF. To measure the back EMF across a phase the conventional method (shown in Fig. 2) requires monitoring the phase terminal and the motor neutral point. The zero crossing of the back EMF can be obtained by comparing the terminal voltage to the neutral point. In most cases, the motor neutral point is not available. The most commonly used method is to build a virtual neutral point that will, in theory, be at the same potential as the neutral point of the wye-wound motor.

This conventional detection scheme is quite simple and has been in use for some time [1]. However, this scheme has its drawbacks. When using PWM to regulate motor speed or torque/current for instance, the virtual neutral point fluctuates at the PWM frequency. As a result there is a very high common-mode voltage and high-frequency noise. Voltage dividers and low-pass filters, as shown in Fig. 2, are required to reduce the common-mode voltage and minimize the high-frequency noise. For instance, if the dc-bus voltage is 300 V, the potential of the neutral point can vary from zero to 300 V. The allowable common mode voltage for a comparator is no where near that level, typically as low as 5 Volts. Obviously, the required voltage divider will reduce the signal sensitivity especially at start-up where it is needed most. Secondly, the required low pass filter will induce a fixed delay independent of rotor speed. As the rotor speed increases, the percentage contribution of the delay to the overall period increases. This delay will disturb current alignment with the back EMF and will cause severe problems for commutation at high speed. Consequently, this method tends to have a narrow speed range and poor startup characteristics.

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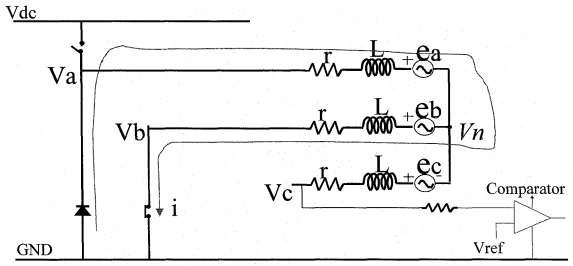


Fig. 3. Back-EMF detection in the PWM off-time moment.

III. Proposed Direct Back-EMF Sensing

As described, the noisy motor neutral point causes problems for the sensorless system. The proposed back-EMF detection successfully avoids using the neutral point voltage when measuring back EMF. The back-EMF voltage can be extracted directly from the motor terminal voltage referred to ground if the proper PWM strategy is selected.

For the two phases being driven the PWM drive signal can be arranged in three ways.

1) *On the high side*, the PWM is applied only on the high side switch, the low side remains on during the step.
2) *On the low side*, the PWM is applied on the low side switch, the high side remains on during the step.
3) *On both sides*, the high side and low side are switched on/off together.

In the proposed scheme, the PWM signal is applied on high-side switches only. The low-side switches are only switched to provide commutation. At any given moment one phase is driven with a PWM high-side switch and another phase is driven with a low-side switch. The remaining phase is left undriven or floating to be measured. The back-EMF signal on the undriven phase is detected during the PWM off time. Fig. 3 shows a circuit model to conduct the analysis.

The voltage on the upper phase during the PWM off time is forced below ground by the motor inductance searching for a current path. At this point, the driven phase voltages are at or just below ground potential. The neutral point potential will be function of each phase's back EMF, and will not be affected by any external driven voltage.

Assuming at a particular step, phase *A* and *B* are conducting current, and phase *C* is floating. The upper switch of phase *A* is pulsewidth modulated to regulate motor speed or current. The lower switch of phase *B* is on during the entire step, switched only at commutation. When the upper transistor of the phase-*A* half bridge is turned off, the current freewheels through the diode *D*. We can calculate what the neutral voltage is during this freewheeling period. From there we can determine the back-EMF voltage on the unused phase *C*.

From the circuit in Fig. 3, it is easy to see $v_c = e_c + v_n$, where V_c is the terminal voltage of the floating phase *C*, e_c is the phase back EMF, and V_n is the neutral voltage of the motor.

From phase *A*, if the forward voltage drop of the diode is ignored, we have

$$v_n = 0 - ri - L\frac{di}{dt} - e_a. \tag{1}$$

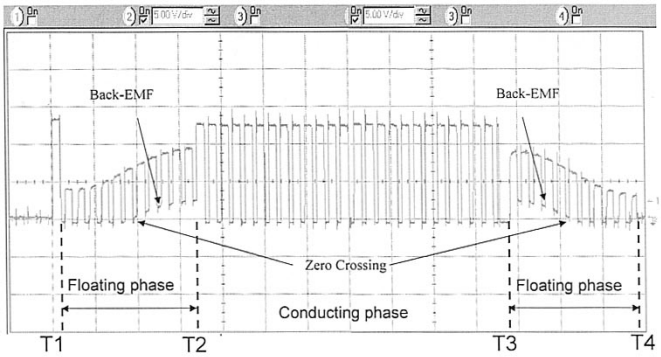


Fig. 4. Phase terminal voltage and the back-EMF waveform.

From phase *B*, if the voltage drop on MOSFET is ignored, we have

$$v_n = ri + L\frac{di}{dt} - e_b. \tag{2}$$

Adding (1) and (2), we get

$$v_n = -\left[\frac{e_a + e_b}{2}\right]. \tag{3}$$

Assuming a balanced three-phase system, we know

$$e_a + e_b + e_c = 0. \tag{4}$$

From (3) and (4),

$$v_n = \frac{e_c}{2}. \tag{5}$$

So, the terminal voltage V_c,

$$v_c = e_c + v_n = \frac{3}{2}e_c. \tag{6}$$

From the above equations, it can be seen that during the off time of the PWM the terminal voltage of the floating phase is directly proportional (with some gain! $1.5e_c$) to the back-EMF voltage without any superimposed switching noise. It is also important to note that this terminal voltage is referenced to the ground with instead of the floating neutral point. So, the neutral point voltage information is not needed to detect the back-EMF zero crossing. As a result there are no common-mode voltage issues. Since the true back EMF is extracted from the motor terminal voltage, the zero crossing of the phase back EMF can be detected very precisely.

Fig. 4 shows a terminal voltage waveform of the circuit. From this waveform, it is clear that the back EMF signal can be extracted from the floating phase terminal voltage. From time *T1* to *T2*, the phase is floating; from time *T2* to *T3*, the phase is being driven; and from time *T3* to *T4*, the phase is floating again. The back-EMF signal can be detected when PWM is "OFF." If the back EMF is negative, it is clamped to about minus 0.7 V by the diode paralleled with the switch in the inverter. When the back EMF is positive, it shows up in the terminal voltage.

Between the time *T1* and *T2*, the rising edge of the zero crossing is detected; and between *T3* and *T4*, the falling edge of the zero crossing can be detected.

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As a summary, several advantages of the proposed-back EMF sensing technique over the conventional schemes can be listed.

- This method has high sensitivity for three reasons. First, since we do not use a voltage divider, so there is no attenuation of the measured EMF. As a result there is still good resolution even at low speed operation. Second, the high frequency switching noise can be rejected because the back EMF is only sampled during the PWM off time. This synchronous sampling can easily eliminate the problems associated with switching noise. Third, this method references the measured back EMF to ground. This eliminates the associated difficulties in dealing with attenuating the signal to accommodate common mode voltage limitations.
- The EMF measurement is not filtered which would cause unwanted delays in the zero-crossing detection. This is especially beneficial at high-speed operation.
- This sensing technique can be easily used in either high-voltage or low-voltage systems without much effort to scale the voltage.
- Faster motor startup is possible due to the precise back-EMF zero-crossing detection without attenuation.
- It is simple and easy to implement, which will be discussed in Section IV.

The synchronous sampling during PWM off time can avoid the switching noise. However, it is mandatory to have a minimum off time (3 μs) to do the sampling. This time is needed for proper settling of the signal prior to sensing the voltage. This translates to a maximum duty cycle that is something less than 100%. What that maximum duty cycle is depends on the chosen PWM frequency.

IV. Hardware Implementation

The proposed back —EMF sensing scheme is implemented in a hardware macro cell inside a microcontroller. This microcontroller integrates the analog detection circuit and other motor control peripherals with a standard micro core, reducing the total system cost. The system block diagram of the sensorless BLDC driver is shown in Fig. 5(b). The motor terminal voltage is directly fed into the microcontroller through current-limiting resistors.

Fig. 5(a) shows the hardware implemented for back-EMF zero-crossing detection. The back-EMF signals go through a multiplexer, where the microcontroller selects which signal to use according to which phase is to be commutated next. Since only zero crossing is of interest, the peak voltage is clamped at 5 V internally. During the PWM off time, the signal is compared to a fixed reference voltage, which is close to zero. The rising edge of the PWM (at the beginning of the PWM on time) will latch the sensed phase comparator output to capture the back-EMF zero-crossing information. Therefore, the sampling happens at the end of the PWM off time. This provides as much time as possible for the turn-off transient noise to settle out. This sampling scheme avoids all switching transients; consequently, all switching noises are rejected.

V. Application Example: Automotive Fuel Pump

The proposed sensorless BLDC drive has been successfully applied to some home appliances for compressors, air blowers,

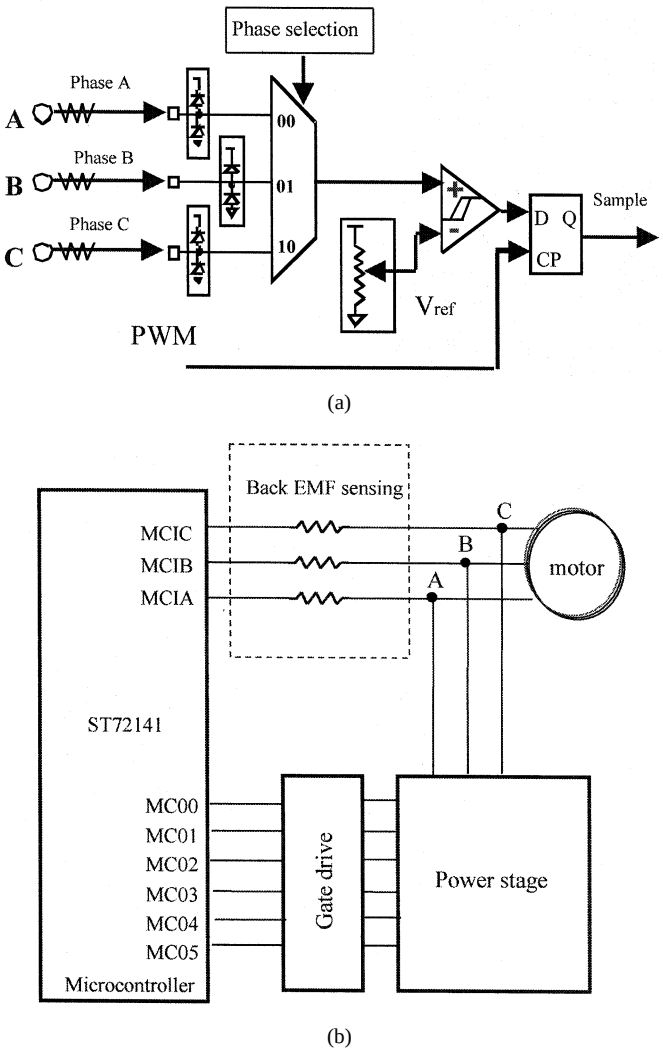


Fig. 5. (a) Back-EMF zero-crossing detection. (b) Block diagram of the proposed microcontroller-based BLDC driver.

and vacuum cleaner applications as well as automotive fuel pump, engine cooling fan, and HVAC blower motor applications. The automotive fuel pump is a good application for this sensorless drive system.

The challenges of the automotive fuel pump application include the following:

1) wide input operating voltage range from 6 to 24 V with the most critical condition at 6-V operation;
2) fast and very reliable startup of the motor within 200 ms;
3) interface compatibility for speed command and communication;
4) low cost.

Thanks to the unique back-EMF sensing scheme and the flexibility brought by the microcontroller, all these issues are solved. Fig. 6 shows the system block diagram for the sensorless drive system.

The battery voltage can vary from 6 to 24 V. At 6 V, the gate driver will not function well. The minimum voltage for the gate driver chip L6387 is around 6.5 V. A supply conditioning circuit, which is a charge pump, is used to boost the voltage for the gate driver. Fig. 7 shows the supply conditioning circuit. The microcontroller will monitor the DC bus voltage (battery voltage).

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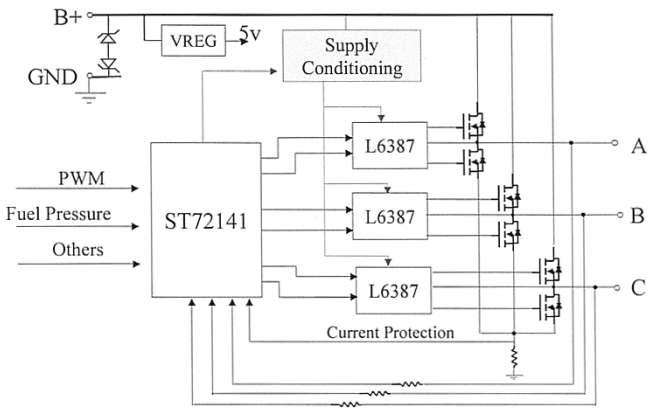


Fig. 6. System block diagram for the sensorless fuel-pump drive system.

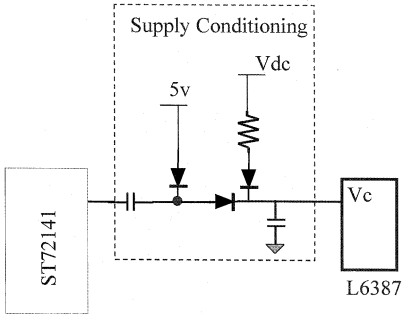


Fig. 7. Supply conditioning circuit for fuel-pump application.

When the dc voltage is below certain threshold voltage (8 V), the charge pump circuit will be activated. This supply conditioning circuit solves the wide voltage range issue.

The second issue is that the motor has to start quickly enough to build up the fuel pressure when the engine is starting. The sensorless BLDC motors are not self-starting. In order to sense the back EMF, the motor must be first started and brought up to a certain speed where the back-EMF voltage can be detected. With the conventional sensorless method, it is very difficult to start the motor very quickly since the motor back EMF is attenuated. Thanks to the proposed high precision back EMF detection of the ST72141, rapid starting of the BLDC motor can be accomplished.

When the motor is not rotating, there is no back EMF to sense for position information. To generate back EMF the controller forces the motor to rotate with open loop commutation. After the motor speed is sufficient to generate enough back EMF to be sensed, the controller switches to synchronous commutation. Since this method actually amplifies the back EMF by 3/2, the motor is able to sense rotation and start sooner than conventional methods. Using this novel back-EMF detection method, it is possible to switch to synchronous commutation when the sensed peak back EMF amplitude is as low as 0.5 V.

In practice, open-loop starting [11] can be accomplished by providing a rotating stator field with a certain increasing frequency profile. When the attraction of the rotor field to the stator field is strong enough to overcome friction and inertia the rotor will turn. After the rotor speed increases to where the back-EMF voltage reaches a detectable level, the position information is available and the system switches to synchronous commutation mode. If there is no specific requirement for startup, like fan applications, this open-loop startup can be satisfactory.

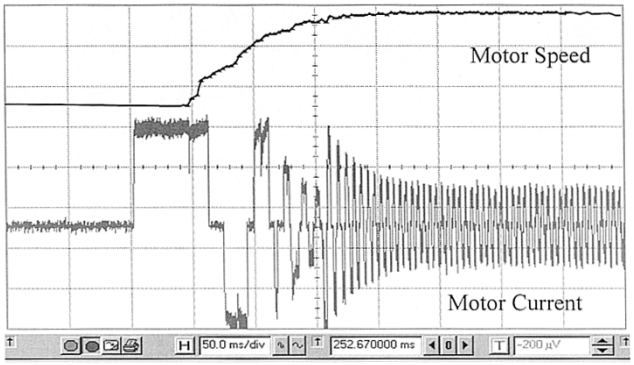


Fig. 8. Startup waveforms of the fuel pump.

For automotive fuel-pump applications, it is desired to have fuel pressure as soon as possible (within 200 ms). This translates to a requirement for maximum acceleration during startup. To start a motor quickly two things must occur. First, start with a known rotor position and second, accelerate the rotor according to the motor inertia and friction parameters. During the software development the maximum acceleration can be determined with help of a tachometer.

The first step is to align the motor to a known position by exciting two phases of the motor. For instance, we can choose phase *A* and phase *B* to be excited to set the initial position. The output of the tachometer will determine when the alignment phase is finished. After the rotor is settled in the initial position, the controller will execute a preset six step excitation pattern. When the motor reaches the maximum speed for each step, the controller should commutate to next step. This will cause the motor to continue accelerating. If the controller commutates too soon or too late maximum acceleration is not achieved. The tachometer can give us the moment of maximum speed, so we can manually set the right value for each step during development. After only a few steps, the back-EMF zero crossing can be detected. Once detected synchronous commutation begins. From that point a closed-loop pressure control algorithm can be implemented.

Fig. 8 shows one phase current and motor speed waveforms during the startup. From the current waveform, the commutation time is indicated. The fuel pump finishes the starting process within 150 ms.

The pump can be controlled either by a pressure sensor directly or through the powertrain control module (PCM). If it is controlled by a pressure sensor directly, the motor speed will be regulated according to the pressure sensor feedback to maintain the desired fuel pressure. If it is controlled by PCM, the fuel pump will receive a speed command. This command may be a PWM pulse train or some other serial communication (LIN, CAN, J1850) from the PCM. This command is typically based on pressure sensor information and expected fuel demand. The motor speed will directly follow the input command from PCM. The microcontroller can easily be programmed to control the motor in either mode.

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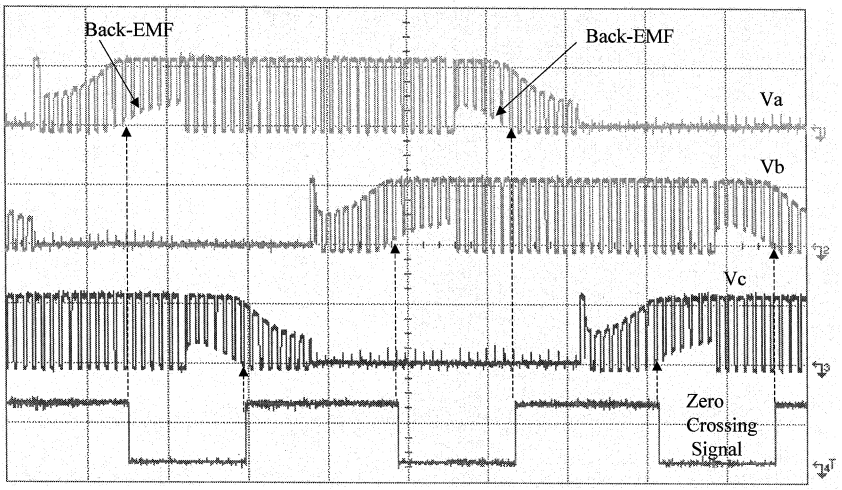


Fig. 9. Three-phase back EMFs and the zero crossings of back EMFs.

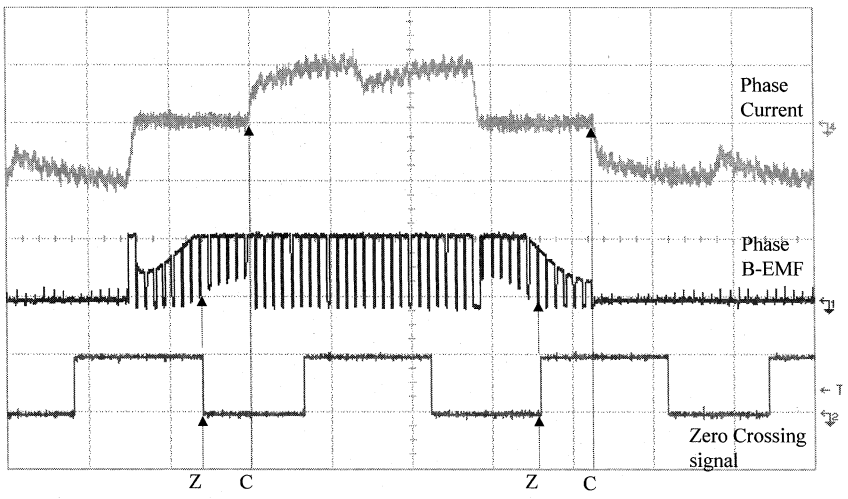


Fig. 10. Sequence of zero crossing of back EMF and phase commutation.

The task of communication is made possible when there is a microcontroller in the system. To take full advantage of the incorporation of a microcontroller in the fuel tank other features can be added at minimal cost. Features such as fuel level, fuel tank pressure, and fuel temperature can be monitored by the microcontroller based module. All this information can be communicated to the PCM through certain protocol. This integration can reduce over all system cost.

The following figures illustrate some key operating waveforms of the sensorless BLDC fuel pump drive system.

Fig. 9 shows the three phase terminal voltages, the back EMFs, and the zero- crossing signal. Each toggling edge of zero-crossing signal corresponds to the zero crossing of the back EMF.

Fig. 10 shows phase back EMF and the phase current. The sequence from back-EMF zero crossing to commutation is clearly demonstrated. Approximately 30 electrical degrees after the zero crossing of the back EMF, the commutation will happen.

VI. Conclusion

A novel back-EMF sensing technique which does not depend on actual or simulated neutral voltage for BLDC drives has been presented in this paper. The true back EMF can be detected in the unused phase during the off time of PWM on the other two phases. This is because the unused phase terminal voltage is directly proportional to that phase's back EMF during this interval. Also, the back-EMF information is referenced to ground without any common mode noise. The high-frequency switching noise is rejected by the synchronous sampling since the back EMF is sampled during the PWM off time. Therefore, this back-EMF sensing method is immune to switching noise. This unique back-EMF sensing method has superior performance to others which rely on neutral voltage information, providing much wider motor speed range with low cost. Furthermore, this back-EMF sensing scheme is embedded in a low-cost microcontroller. This microcontroller integrates the analog detection circuit and other peripherals for motor control with a standard micro core, reducing the total system cost.

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The test results verify the analysis and demonstrate the advantages of this novel back-EMF detection scheme and mixed-signal microcontroller system. The proposed sensorless BLDC drive system has been successfully applied to automotive fuel pump applications. Sensorless BLDC applications can include hard disk drives, fans, pumps, blowers, and home appliances as well.

REFERENCES

[1] GE, "Control system, method of operating an electronically commutated motor, and laundering apparatus," U.S. Patent 654 566, 1987.

[2] T. Endo and F. Tajima, "Microcomputer controlled brushless motor without a shaft mounted position sensor," in *Proc. IPEC-Tokyo*, 1983, pp. 1339–1345.

[3] K. Rajashekara, A. Kawamura, and K. Matsuse, *Sensorless Control of AC Motor Drives*. New York: IEEE Press, 1996.

[4] R. Becerra, T. Jahns, and M. Ehsani, "Four quadrant sensorless brushless ECM drive," in *Proc. IEEE APEC'91*, 1991, pp. 202–209.

[5] J. Moreira, "Indirect sensing for rotor flux position of permanent magnet AC motors operating in a wide speed range," in *Conf. Rec. IEEE-IAS Annu. Meeting*, 1994, pp. 401–407.

[6] S. Ogasawara and H. Akagi, "An approach to position sensorless drive for brushless dc motors," *IEEE Trans. Ind. Applicat.*, vol. 27, pp. 928–933, Sept./Oct. 1991.

[7] STMicroelectronics, "Control of a brushless motor," U.S. Patent 5 859 520, 1999.

[8] (2001) An Introduction to Sensorless Brushless DC Motor Drive Applications With the ST72141 (STMicroelectronics Application Note AN1130). STMicroelectronics. [Online]. Available: www.st.com

[9] J. Shao, D. Nolan, and T. Hopkins, "A novel direct back EMF detection for sensorless brushless DC (BLDC) motor drives," in *Proc. IEEE APEC*, 2002, pp. 33–38.

[10] J. Johnson, "Review of sensorless methods for brushless DC," in *Conf. Rec. IEEE-IAS Annu. Meeting*, 1999, pp. 143–150.

[11] R. Krishnan and R. Ghosh, "Starting algorithm and performance of a PM DC brushless motor drive system with no position sensor," in *Proc. IEEE PESC'89*, 1989, pp. 815–821.



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He is currently with STMicroelectronics, Schaumberg, IL. He has done extensive design and development work in control software and circuitry for brush dc, sine, and six-step commutated brushless dc (sensored and sensorless), and induction motors using phase-controlled SCR rectifiers and PWM-controlled transistor inverters. He is the holder of seven U.S. patents in the motor control field.



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In 1996, he began his career with STMicroelectonics in the European industrial application laboratory in Munich, Germany. He then spent four years as a Field Application Engineer in Boston, MA, supporting American microcontroller customers. He recently moved to Rousset, France, in the Microcontroller Division, where he will be responsible for automotive-targeted microcontrollers definition and technical support.



David Swanson received the B.S.E.E. degree (*magna cum laude*) from North Carolina State University, Raleigh, in 1985.

He has been around the automotive industry most of his life. After obtaining a diploma in auto mechanics, he was a mechanic for three and one-half years prior to starting college. He started out working for Delco Products, Dayton OH, as a Production Electronics Engineer, where he worked on various systems from brushless motors to ultrasonic height sensors. He left Delco for STMicroelectronics (then SGS) in 1987 where he filled the roles of Technical Support Engineer, Market Development Manager, Design Center Manager, and now Staff Engineer in Livonia, MI. He is the holder of several automotive-related from instrumentation to charging systems.

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